

Humanities (History & Geography)

Year 9

Cambridge Lower Secondary

Student Manual



Prime Books · Humanities (History & Geography) Year 9 · Student Book

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British English edition.

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INTRODUCTION

How this book works

Everything you build, grow and drink depends on the rock underneath. In this first unit you will identify the three rock groups, follow a real Portuguese marble from seabed to quarry, and discover how rock decides the shape of a landscape and the fertility of its soil.

What makes this book different

- **Everything is original.** Every sentence, example and diagram was made for this book.
- **Every claim is checked.** Facts come from real, cited sources; every numerical answer was executed by computer before publication.
- **Thinking is taught explicitly.** Dedicated panels show you how specialists in this subject actually reason.

A note to teachers and parents

This book follows the spirit of an international lower-secondary programme, interpreted afresh for Prime School. Each topic opens with clear outcomes and closes with self-assessment, so progress is visible to pupil and teacher alike.

HOW TO USE THIS BOOK

Reading the page

Each section follows the same rhythm, so you always know where you are.

PANEL	WHAT IT DOES
Learning outcomes	What you will be able to do by the end.
Getting started	A warm-up that wakes up what you know.
Key words	The vocabulary of the topic, defined.
Source	A photograph, map or document to study.
Your turn	Questions and activities for you to do.
Think like a geographer	A task that builds geographical thinking.
Summary checklist	'I can' statements to check your progress.

How to work

Read with a pencil in your hand. When the book asks a question, try to answer it before reading on. Do the work in your notebook, and check your reasoning, not just your answer.

GETTING SET UP

Before Unit 1

You need a pencil, a ruler, a notebook and your curiosity. Anything else this unit requires is listed where it is needed.

Your toolkit

- A notebook for working, planning and recording, not just for final answers.
- The habit of checking: recalculate, reread, and ask whether the answer is sensible.
- The confidence to say "I am not sure yet", which is where all real learning starts.

A quick self-check

- 1 What do you already know about this topic?
- 2 Which part of it do you expect to find hardest, and why?
- 3 How will you know when you have genuinely understood something, rather than just recognised it?

Unit 1 begins now. Work carefully, check honestly, and ask good questions.



UNIT 1

1

From rock to soil

The stone beneath your feet, and what becomes of it

© IN THIS UNIT, YOU WILL

- ✓ Explain why rock matters to how people live.
- ✓ Identify the three rock groups and how each forms.
- ✓ Describe weathering and the rock cycle.
- ✓ Explain how rock shapes relief and soil.

◆ GETTING STARTED

Everything you build, grow and drink depends on the rock underneath. In pairs, discuss:

- what is your house built from, and where did that material come from?
- why do you think some regions have quarries and others have farmland?
- what do you think soil actually is, and where did it come from?

UNIT 1 • TOPIC 1.1

Our rocky home

Learn

Rock decides a surprising amount about human life. It decides which land farms well and which does not, where water collects and where it drains away, what buildings are made of, and which places have industries built on stone.

Portugal is a good place to see this. The granite north builds with grey stone and has thin, acidic soils. The limestone belt around Leiria holds water in underground caves. The Alentejo has marble so fine that it has been quarried for two thousand years.

● YOUR TURN

- 1 Give three ways the rock beneath a place affects the people living there.
- 2 Why might two farms twenty kilometres apart grow completely different crops?
- 3 Look around your classroom. List three things made from rock or from materials that came out of the ground.

UNIT 1 • TOPIC 1.2

The three rock groups

Learn

All rocks belong to one of three groups, defined by **how they formed**.

GROUP	HOW IT FORMS	EXAMPLES
igneous	Molten rock cools and hardens, either underground (slowly, large crystals) or at the surface (quickly, small crystals)	granite, basalt
sedimentary	Sediments settle in layers, are buried and compacted over millions of years	limestone, sandstone, shale
metamorphic	Existing rock is changed by great heat and pressure, without melting	marble, slate, schist

“ **SOURCE A · ESTREMOZ MARBLE, ALENTEJO**

Around 370 million years ago, southern Portugal lay beneath a warm shallow sea. Shells and lime mud settled on the seabed and hardened into limestone, a sedimentary rock. Later, the collision that built the Variscan mountains subjected that limestone to intense heat and pressure. It did not melt. Its crystals grew and reorganised, and it became **marble**, a metamorphic rock.

Today the Estremoz Anticline, a folded structure about 40 km long and 8 km wide between Estremoz, Borba and Vila Viçosa, is Portugal's main marble district. The stone has been quarried since Roman times and is used in buildings across the world.



Figure 1.1 Marble quarries in the Alentejo, cut into a folded anticline.

• YOUR TURN

- 1 Name the three rock groups and explain how each one forms.
- 2 Using Source A: a. What rock was Estremoz marble before it was marble?
a. What changed it? c. Why is it metamorphic and not sedimentary?
- 3 Granite cooled slowly underground and basalt cooled quickly at the surface. Which do you think has larger crystals? Explain your reasoning.
- 4 Suggest why marble quarrying became an industry in the Alentejo rather than somewhere else in Portugal.

Weathering

Learn

Weathering is the breakdown of rock **in place**, without being carried away. When the broken material is transported, that is **erosion**. Geographers are careful about the difference.

There are three kinds of weathering.

- **Physical (mechanical):** rock is broken apart without changing chemically.

Water freezes in a crack, expands, and levers the rock open. Repeated heating and cooling makes surfaces flake.

- **Chemical:** rainwater is slightly acidic, and dissolves some minerals.

Limestone and marble are especially vulnerable, which is why old gravestones lose their lettering.

- **Biological:** living things break rock down. Tree roots widen cracks; lichens

release acids that eat into the surface.

- **YOUR TURN**

- 1 Explain the difference between weathering and erosion.
- 2 Name the three types of weathering and give an example of each.
- 3 Why is freeze-thaw weathering more common in mountains such as the Serra da Estrela than along the Algarve coast?
- 4 Marble is beautiful but chemically weathers faster than granite. What problem does this create for marble statues in cities?

The rock cycle

Learn

Rock is not permanent. Over immense timescales, every rock is destroyed and remade. This is the **rock cycle**.

Weathering breaks rock into sediment. Rivers, wind and ice **transport** it. It is **deposited**, buried and compacted into sedimentary rock. Pushed deeper, heat and pressure turn it **metamorphic**. Deeper still, it melts to **magma**, which cools into **igneous** rock, is uplifted to the surface, and weathers again.

There is no start and no end. Estremoz marble is one loop of this cycle caught mid-turn: seabed to limestone to marble, and one day, sediment again.

- YOUR TURN

- 1 Draw the rock cycle as a labelled diagram, showing all three rock groups.
- 2 Describe the full journey of a grain of sand from a granite mountain to a metamorphic rock.
- 3 Explain why geographers say rock is "recycled" rather than "used up".

UNIT 1 • TOPIC 1.5

Rock, relief and soil

Learn

Relief means the shape and height of the land. Hard rock resists weathering and stands out as hills and cliffs. Softer rock wears away into valleys and plains. The shape of a landscape is largely a map of rock hardness.

Soil is the thin skin where geology meets life. It forms from weathered rock mixed with **humus**, the decayed remains of plants and animals, plus water, air and countless organisms. It can take **hundreds of years to form one centimetre** and minutes to lose it to a bulldozer or a flood.

Rock type shapes soil directly. Granite gives thin, sandy, acidic soils. Limestone gives thin, alkaline soils that drain fast. River floodplains give deep, fertile soils, which is why people have always farmed them.

- YOUR TURN

- 1 What is relief, and how does rock hardness affect it?
- 2 What is soil made of? Name four components.
- 3 Why is soil described as a non-renewable resource on a human timescale?
- 4 Explain why the soils of a river floodplain are usually more fertile than those on a granite hillside.

☒ SUMMARY CHECKLIST

- ✓ I can explain how rock affects the way people live.
- ✓ I can identify the three rock groups and how each forms.
- ✓ I can distinguish weathering from erosion and describe three types.
- ✓ I can describe the rock cycle.
- ✓ I can explain how rock shapes relief and soil.

UNIT 1 • FINAL PROJECT

A geological profile of your area

★ FINAL PROJECT

A geological profile of your area

Investigate the rock beneath your own area and produce a geological profile explaining how it shapes the landscape, the soil and the way people live there.

What to hand in

- A named identification of the local rock, with its group and how it formed.
- A labelled cross-section from bedrock to topsoil.
- An explanation of how that rock affects local relief, soil and land use.
- A sources list, with links, for every fact you used.

How it will be judged

Criterion	Weight	--- ---	Correct geology and rock identification	30%	Quality of the cross-section	25%	
Explanation linking rock to human activity				30%	Referencing	15%	

UNIT 1 • CHECK YOUR PROGRESS

- 1 Name the three rock groups and explain how each forms. [3]
- 2 Estremoz marble began as limestone. Explain how it became marble. [2]
- 3 Explain the difference between weathering and erosion. [2]
- 4 Name the three types of weathering and give an example of each. [3]
- 5 Describe the journey of a sand grain from granite mountain to metamorphic rock. [3]
- 6 What is relief, and how does rock hardness affect it? [2]
- 7 Why is soil described as non-renewable on a human timescale? [2]

UNIT 1 • EVALUATION

Be honest. Colour the circle that matches how you feel about each outcome: green means confident, amber means nearly there, red means needs more work.

I CAN ...	GREEN	AMBER	RED
explain how rock affects the way people live	☐	☐	☐
identify the three rock groups and how each forms	☐	☐	☐
distinguish weathering from erosion	☐	☐	☐
describe the rock cycle	☐	☐	☐
explain how rock shapes relief and soil	☐	☐	☐

UNIT 1 • WHAT CAN YOU DO?

✓ WHAT CAN YOU DO?

- | | |
|--|---|
| ☐ Explain how rock affects the way people live. | ☐ Describe the rock cycle. |
| ☐ Identify the three rock groups and how each forms. | ☐ Explain how rock shapes relief and soil. |
| ☐ Distinguish weathering from erosion and describe three types. | |

GLOSSARY

Unit 1

igneous: rock formed when molten rock cools and hardens.

sedimentary: rock formed from compacted layers of sediment.

metamorphic: rock changed by heat and pressure without melting.

weathering: the breakdown of rock in place.

erosion: the transport of weathered material away.

rock cycle: the endless destruction and reforming of rock.

relief: the shape and height of the land.

humus: the decayed plant and animal matter in soil.

SOURCES & REFERENCES

Where everything came from

All text and diagrams are original to this edition. The factual claims can be checked here. Links were live as of July 2026.

§ UNIT 1 • FROM ROCK TO SOIL

- 1 **Estremoz marble, Alentejo. Paleozoic calcitic marble of the Estremoz Anticline, about 40 km long and 8 km wide, metamorphosed from limestone under green-schist conditions.**

<https://repository.europe-geology.eu/egdidocs/eurolithos/eurolithos+atlas+portugal.pdf>

- 2 **The origin of the marble. Southern Portugal lay beneath a warm shallow sea over 370 million years ago, forming the limestone that later became marble.**

https://www.researchgate.net/publication/269756334_Global_Heritage_Stone_Estremoz_Marbles_Portugal

- 3 **The rock cycle and weathering. The processes by which rock is broken down and reformed.**

https://en.wikipedia.org/wiki/Rock_cycle

The ground beneath everything.

This Year 9 Student Book starts at the bedrock. You will identify igneous, sedimentary and metamorphic rock, trace Estremoz marble from a shallow sea to an Alentejo quarry, distinguish weathering from erosion, and explain how rock shapes relief and soil.

Written for Prime School pupils, in clear British English, with real places and cited sources.

ROCKS

WEATHERING

ROCK CYCLE

RELIEF

SOIL